Exploring the correlation between Be star rotation axis direction and galactic features

Final Progress Report (CS4490Z)

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April 8, 2024

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Glossary

Be star: A special type of B-type star with a disk of gas expelled from its equator. These disks are formed by high-speed rotation that causes the star's material to be thrown outward.

Axis of Rotation: An imaginary line around which a star or other celestial body rotates around itself.

Inclination angle: The angle between the axis of rotation and the observer's line of sight.

Position angle: The angle between the north direction and the axis of rotation's projection on the sky plane.

Galactic Plane: The central plane or disk of the Milky Way, in which most of the Milky Way's stars and other matter are concentrated.

Monte Carlo Simulation: A method of solving computational problems using random numbers (or more broadly, random variables), used in this study to construct synthetic data sets with specified levels of correlation.

Synthetic Data Sets: A data set generated by simulation or other computational methods for analysis or testing rather than collected directly from actual observations.

Correlation: A measure of interdependence between two or more variables, used to study the statistical relationship between them.

Alpha (α): A parameter set in the Monte Carlo simulation to control the weight of the preferred direction's influence on the direction of the simulated star's rotation axis.

Preferred Direction ($\vec{p}$ vector): The vector used to specify a preferred direction for the stellar's rotation axes. Simulations can assume a correlation between 0 and 100%with the direction.

Python: A widely used high-level programming language commonly used in the fields of data science, machine learning, and astronomy.

Astrophysics: The branch of science that studies the physical properties and behavior of celestial bodies in the universe.

Milky Way: Our Galaxy which contains the Sun, Earth, and tens of billions of other stars, as well as large amounts of interstellar gas and dust.

Chi-squared Test (χ^2 Test): A statistical test used to evaluate whether two sets of data are significantly different, or the differences can be ascribed to chance alone.

Kolmogorov–Smirnov Test (KS Test): A non-parametric test used to determine whether two samples come from the same distribution, or whether a sample comes from a specific distribution.

Structured Abstract

Context and motivation

A Be star's disc allows one to extract the direction of its axis of rotation. The angle between the rotation axis and the line of sight, the inclination angle, can be extracted from the spectrum. The projection of the axis of rotation onto the sky plane can be determined by polarization measurements and therefore the position angle can be determined.

Research Question

Explore correlations in the orientation of the rotational axis between Be stars or between Be stars and galactic features such as the galactic plane.

Principal ideas

Use Monte Carlo simulation to construct synthetic data sets with specified correlation levels. Synthetic and real datasets are compared to obtain the best representative correlations and preferred directions.

Research Methodology

This research is mainly data science research and will develop a Python software package that can analyze one or more data sets and extract the correlation level between the Be star rotation axes from these data. If the program finds a significant correlation, its preferred direction in space will be extracted.

Anticipated results

Expected results include correlation results of Be stars with galactic features and a Python package for statistical analysis of the data.

Anticipated Novelty

The unique properties of Be stars will allow more precise determination of stellar orientations and reveal their correlation with the galactic plane. Any statistically significant correlation will be a publishable result.

Anticipated impact of results

These findings could have a significant impact on theoretical astrophysics and practical methods of stellar study, providing new insights into star formation of the Milky Way and in open stellar clusters.

Limitation

Potential limitations include inaccuracies in existing data on Be stars and the inherent complexity of accurately interpreting astronomical data. Furthermore, the statistical methods employed may have inherent limitations in establishing causality and correlation.

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Introduction

In the field of astrophysics, the study of the direction of stellar rotation axes is crucial for understanding the properties of stars themselves and their distribution patterns in the Milky Way. In particular, Be stars, a special class of B-type stars that form a distinct periastron disc near the equator due to their rapid rotation, can accurately obtain the direction of the rotation axis, making them ideal objects for the study of the direction of the stellar rotation axis. This study aims to delve into the potential correlation between the rotational axis directions of Be stars and the large-scale structure of the Milky Way in an attempt to reveal whether there is a trend towards unification between the rotational axis directions of these stars and the characteristics of the Milky Way.

This paper focuses on whether the direction of Be stars' rotational axes tends to show some correlation with Galactic features. By constructing synthetic datasets containing specified correlation levels through Monte Carlo simulation methods and comparing the distributional differences between the synthetic and the actual datasets using statistical analysis methods, we attempt to extract the correlation and preference direction of the rotational axis orientations from them.

The main results of the study reveal that, to some extent, the rotational axis directions of Be stars do exhibit a correlation with the Galactic plane. The recovered preferred directions show a clear tendency for the rotational axis directions of Be stars to be more perpendicular to the Galactic plane. We find that our method has a high accuracy in recovering theta values for the rotational axis directions, while there are some limitations for the phi values.

The structure of this report is organized as follows: firstly, we review the research background and related work, then we clarify the research objectives and methodology adopted, followed by a detailed presentation of the results and discussion, and finally, we summarize the conclusions, plan the direction of future work and summarize the lessons learned.

Background and Related Work

Scientific and technological development

The study of Milky Way Be stars represents an important aspect of current astrophysics research. Be stars are faster-rotating B-type stars characterized by main-sequence stars with an equatorial circumstellar disk. The unique circumstellar gas disk of Be stars makes it easier to confirm the direction of the rotation axis, providing a unique opportunity to explore its potential correlation with galactic features such as the galactic plane.

The position angle of the Be star's disk can be determined by detecting the polarization of the Be star's light. Normally light from stars is unpolarized, meaning the electric fields oscillate in random directions. However, when the light interacts with the gas in the Be star's circumferential disk, it becomes polarized, meaning the electric field oscillates mainly in one plane. By measuring the angle of this plane relative to some reference direction, the position angle of the disk can be determined (Yudin, 2001).

Recent advances in determining Be star tilt are driven by advances in machine learning techniques in spectroscopic techniques. As shown in Figure 1, the inclination angle i of a star is the angle between the star's rotation axis and the observer's line of sight. A study by Sigut and Ghafourian showed that using Hα emission line profile fitting has proven effective in determining the inclination of Be stars with considerable accuracy (Sigut & Ghafourian, 2023). This method supports the extraction of equatorial rotation speed and is beneficial to the study of stellar spin alignment. In addition, Sigut and Lailey used machine learning to identify the Be star tilt angle in the spectrogram and found that the tilt angle error calculated by the regression neural network algorithm on the observation sample was very small (Lailey & Sigut, 2023). This method quickly and efficiently determines the inclination of observed Be stars, facilitates searches for relevant spin axes in young open clusters, and extracts equatorial rotation velocities from measurements.

Figure 1

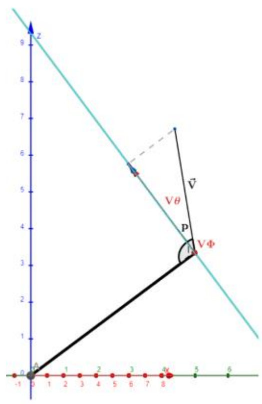
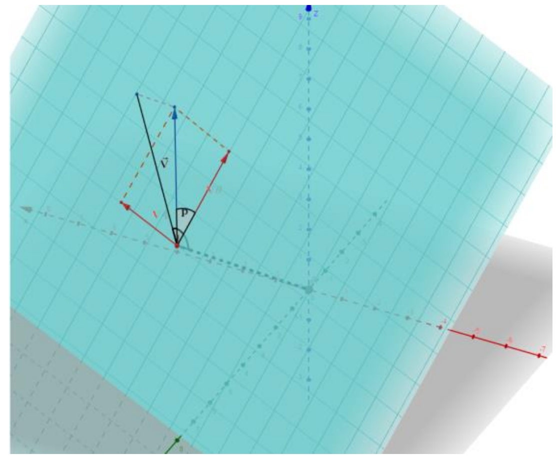


Figure 2

Figure 3



Note. I in Figure 1 represents the Inclination angle, which is the angle between the star's rotation axis V and the line connecting the star's coordinates to the observer's position.

P in Figure 2 represents the Position angle, which is the angle between the star's rotation axis V and the north component of the projection of V on the sky plane.

Figure 3 shows the overall view, where the cyan plane is the sky plane, perpendicular to the line connecting the star coordinates to the observer's position.

Research gaps

A current research gap lies in the limited understanding of the relationship between the axis of stellar rotation and the larger galactic structure. Helmut A. Abt used the predicted rotation speed and rotation period to infer the direction of the rotation axis of 102 Ap stars, stars with chemical spots, and concluded that within the error range, the rotation axis of the Ap stars was randomly oriented (Abt, 2001). Although no correlation was found between the axis of stellar rotation and the latitude and longitude of the Milky Way, Abt set a precedent for such studies. In addition, the method used by Abt had large errors. For example, many stars had $sin\ i > 1$ which is impossible.

Aiming to understand the evolution of angular momentum during the formation of star clusters, Corsaro's team used asteroseismology to study the spin arrangement of stars in open clusters (Corsaro et al., 2017). By measuring the spin axis inclinations of the stars in NGC 6791 and NGC 6819, Corsaro's team found that the spin axes within each cluster are strongly aligned: about 70% of the stars show low to medium tilt alignments (Corsaro et al., 2017). The results of this study demonstrate an efficient transfer of global angular momentum from the cluster-forming clouds to individual stars, demonstrating that a significant portion of the protocluster's initial kinetic energy requires rotation to achieve the observed levels of stellar spin alignment.

The Curé team used Monte Carlo simulations to reveal a bimodal distribution of Be star projection polarization angles, challenging the initial assumption of uniform distribution (Curé et al., 2010). Research shows that due to the influence of the shape of the galaxy, the distribution of the rotation axis of Be stars is not uniform (Curé et al., 2010). The uneven distribution of the intrinsic polarization angles of Be stars means there may be a correlation with other factors.

The foundation has been laid by studies of Ap stars and Enrico Corsaro's study of spin alignment in ancient open clusters, but a comprehensive analysis is still needed specifically for Be stars and their alignment with the galactic plane. This study aims to bridge this gap by leveraging data science tools to visualize and analyze the Be star cluster associated with the Milky Way.

Research Objectives

01: **Construct star simulation data**: Use Monte Carlo simulation to create 1000 stars with random coordinates in a cylindrical space, and generate a rotation axis vector for each star to explore the distribution of star rotation axis orientations feature.

02: **Simulation of rotation axes preference direction**: Important parameters for each simulation are vector $\vec{p}$ giving the preferred direction for the rotation axes and α [0,1], giving the statistical correlation between the stellar rotation axes and $\vec{p}$.

03: **Construct Database**: Generate a simulation database with different α values and $\vec{p}$ vectors, and save relevant statistical data to restore relevant parameters.

04: **Test random sample**: Through the chi-square test or KS test, use random samples and simulated databases to test the effect of recovering the $\vec{p}$ vector and related parameters from the target sample.

05: **Compare real star data and simulation results**: Use real Be star data to compare with the simulation database, analyze the frequency array of inclination angle and position angle through the chi-square test or KS test, and find the simulation with the highest distribution correlation data.

06: **Extract evidence of a unified rotation axis orientation**: Based on the degree of matching between simulation and real data, evaluate whether there is a unified rotation axis orientation on the Be star.

07: **Evaluate limitations of research methods**: Clearly identify and discuss potential limitations of the data and statistical methods used in this study and how these limitations may affect the interpretation of the results.

Methodology

Research design

This study aims to explore the correlation between the direction of the rotation axis of Be stars and galactic features such as the galactic plane. The main research method adopted is data science research, by developing a Python software package that processes one or more star data sets to extract the level of correlation between the rotation axes of Be stars. If a significant correlation is found, its preferred direction in space will be further extracted.

Data collection

To generate the synthetic data set, this study uses the Monte Carlo simulation method, and the parameters of the simulation include the spatial distribution of the stars and the direction of the rotation axes. By adjusting the alpha value (α) and the preference direction vector ($\vec{p}$ vector), simulate different rotation axis distributions. The rotation axis vector of each star will be created using the formula $(1-\alpha)\vec{V} + \alpha \cdot \vec{p}$, where $\vec{V}$ is a random vector. By changing the α value, we can simulate the extent to which the star's rotation axis is oriented in the preferred direction.

We will collect real data about the Be star from databases and published literature. These data mainly include the position angle, inclination angle and other related astronomical observation data of the Be star. Selection criteria will be based on data reliability, completeness and recent observations.

Data processing and analysis

The developed Python software package can simulate or import real stellar data sets, process the data by applying statistical analysis methods, and extract correlations and preferred directions between rotation axes. The chi-square test and KS test will be used to compare the distribution of inclination angles and position angles of the real data set and the simulated data set to determine the correlation between the two. These statistical tests help identify significant distribution differences and potential uniform directionality.

Verification and testing of results

Simulated star test: When $\vec{p}$ vector is not used to generate a simulated star data set, verify whether the distribution of inclination angle and position angle agrees with the expected $sin\ i$ distribution and uniform distribution, respectively.

Alpha value recovery test: After trying to recover the α value from the database, verify whether our analysis method can accurately recover the specific α value by comparing it with the simulated data of the preset α value.

Test of $\vec{p}$ vector: When the α value is high, verify whether the set $\vec{p}$ vector can be accurately identified and check its consistency with the actual data.

Tests in Galaxy Mode and Cluster Mode: Two different observation modes will be tested: one is Galaxy mode (the observer is at the center of the Galaxy), and the other is Cluster mode (the observer is far away from the star cluster).

Research limitations

This study may be limited by the accuracy of existing Be star data and the difficulty of interpreting astronomical data. In addition, the statistical methods used have limitations in establishing causality and correlation, which may affect the interpretation of the results.

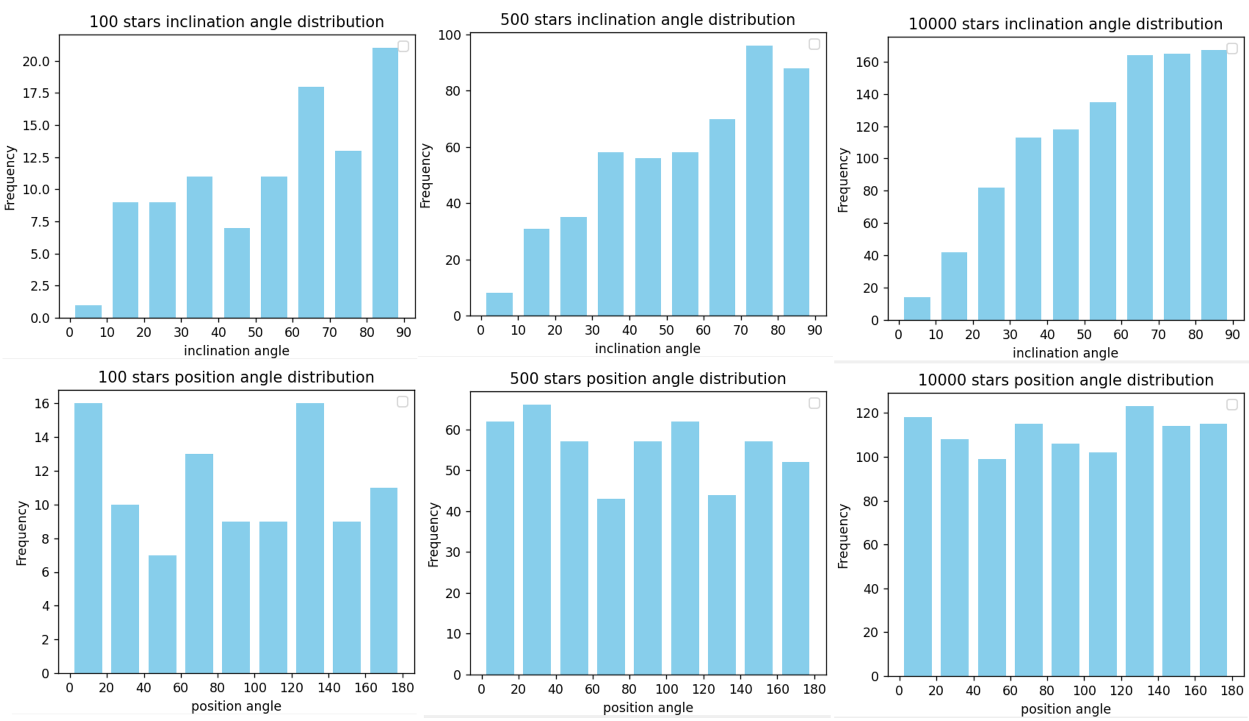
Results

Data Collection

This study uses the Monte Carlo simulation method to generate a simulation data set, aiming to simulate the distribution of the Be star's rotation axis direction. First, set the basic parameters of the simulation process, including the preferred direction vector ($\vec{p}$ vector) and correlation parameter (α value). By mixing randomly generated vectors and preset $\vec{p}$ vectors, the direction of the rotation axis of each star is simulated. In this process, the α value is used to adjust the weight between the random vector and the $\vec{p}$ vector to explore the impact of different preference directions on the rotation axis distribution.

When the α value is 0, the inclination angle distribution of the generated star data set will be approximated by the $sin\ i$ distribution, and the position angle distribution will be approximated by the uniform distribution. Figure 4 shows that when the number of stars in the data set increases, the distribution similarity becomes higher.

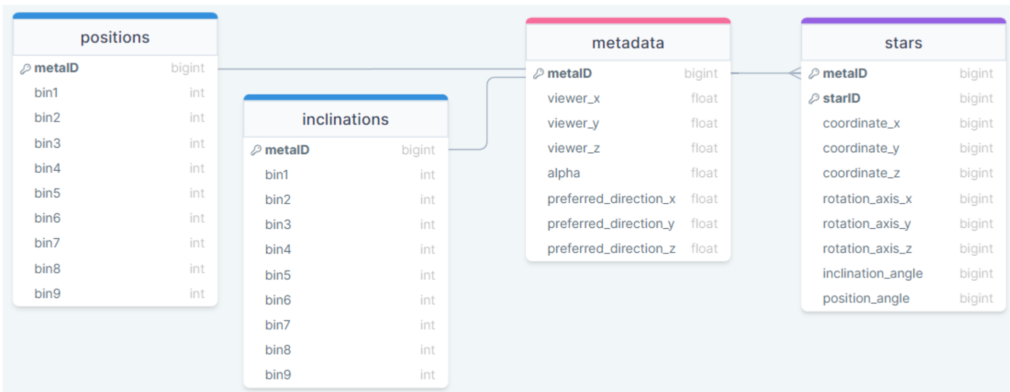
Figure 4



Note. Figure 4 shows the distribution of inclination angle and position angle in data sets with different numbers of stars.

Based on the above simulation process, we constructed a database containing simulated Be star rotation axis direction data. Figure 5 shows the structure diagram of the database. The stars table saves each star’s data. Each star has random coordinates. The rotation axis vector is generated based on different α values and $\vec{p}$ vectors, and the inclination and position angle are calculated using the rotation axis vector and coordinates. The metadata table stores the viewer coordinates, α value and $\vec{p}$ vector used to generate each star data set. In this database, a star data set holds 1000 stars. The inclinations table and the positions table respectively store the frequency arrays of the inclination angle and position angle of each star data set.

Figure 5



Note. Figure 5 shows the structure diagram of the database

To comprehensively analyze the distribution characteristics of the rotation axis direction, the database records star data sets generated under different combinations of α values and $\vec{p}$ vectors. The generated database uses 11 α values between 0 and 1 and 12,604 $\vec{p}$ vectors evenly distributed within the sky sphere. Each data set is generated by a $\vec{p}$ vector and a α value, and there are 138,644 data sets in the database.

The structure of the database is designed to facilitate query and analysis, supporting multi-angle statistical analysis of simulated data sets. By setting different query conditions, data subsets under specific parameter combinations can be easily extracted to facilitate further data analysis and testing.

Database testing

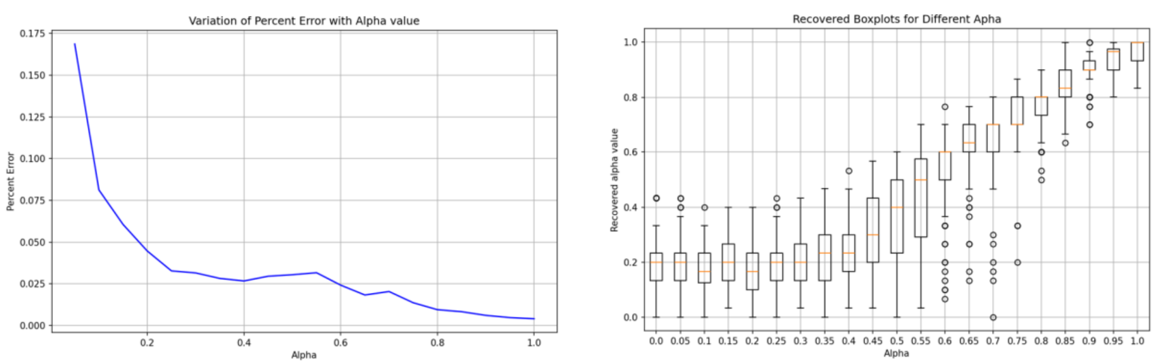
The database test part focuses on verifying the accuracy and effectiveness of the model in restoring the preset α value and $\vec{p}$ vector. By designing a series of test cases, each use case targets a specific α value and $\vec{p}$ vector configuration to generate a corresponding simulated star data set. Subsequently, using the developed search tool, attempts were made to recover these preset parameters from the database to evaluate the recovery capability of the database.

After using the α value and $\vec{p}$ vector to generate simulated star samples, to find data matching these samples from the database, we compared the frequency arrays of the sample's inclination angle and position angle with the corresponding frequency arrays in the database. We used reduced chi-square test as a statistical method to determine which database sample has the highest consistency with the target sample. Through this method, we can evaluate the similarity between each simulated sample in the database and the observed sample, and then recover the αvalue and $\vec{p}$ vector from the database.

Different levels of α values were used during the test, ranging from completely random ($\alpha = 0$) to completely biased towards the $\vec{p}$ vector ($\alpha = 1$), to comprehensively evaluate the performance of the database in various extreme and intermediate states. At the same time, the $\vec{p}$ vector is also set to multiple directions to test whether the database can accurately identify the preferred trend of the direction of the star's rotation axis. After each test, the success of the test is evaluated by calculating the deviation between the actual recovered parameters and the preset parameters. The size of the deviation directly reflects the accuracy of the database in parameter recovery.

We paid particular attention to the accuracy of database recovery α values. As shown in Figure 6, when we use randomly selected $\vec{p}$ vectors and specific α values to generate different star data sets, we find that as the α value increases, the accuracy of the recovery also increases. Specifically, when the α value is higher than 0.6, although there are still some outliers, it can be seen from the decrease in the percentage error and the central tendency of the box plot that the error is significantly reduced. This trend indicates that our database exhibits higher recovery accuracy when handling higher α values.

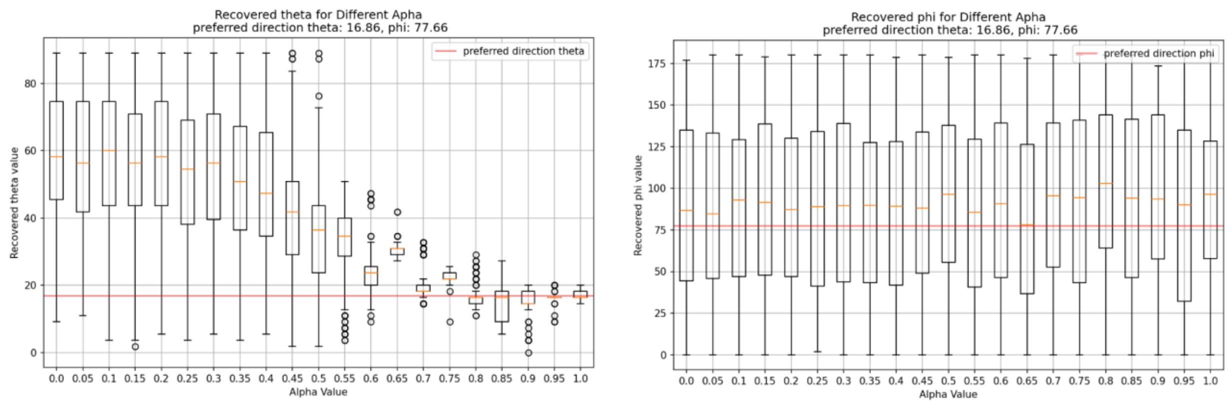
Figure 6



Note. Figure 6 shows the accuracy of the α values recovered from the database as the α value changes.

We further explored the performance of recovering the $\vec{p}$ vector under different α values. The $\vec{p}$ vector is represented by the spherical coordinate system and is determined by the two angles theta (θ) and phi (φ) respectively. As shown in Figure 7, as the α value increases, especially when the α value is greater than 0.6, the recovery θ becomes more accurate, which shows that the model exhibits high accuracy in recovering the latitude angle of the $\vec{p}$ vector. The red line marks the actual θ value and φ value of the $\vec{p}$ vector. We notice that the θ value tends to be stable as the α value increases, while the φ value remains relatively consistent throughout the α value range and does not show any change with the α value. Increasing and concentrated tendencies. This indicates that there are certain challenges in recovering the longitude angle of the $\vec{p}$ vector in the database.

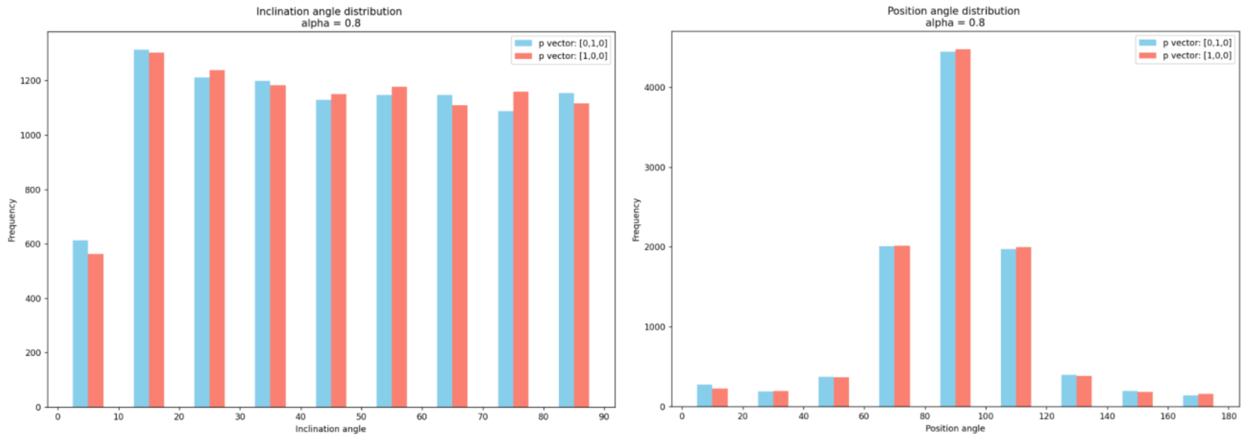
Figure 7



Note. Figure 7 shows the test results of using the database to restore the $\vec{p}$ vector.

The inability to accurately recover φ values reveals a limitation of our simulation approach. As shown in Figure 8, two mutually perpendicular $\vec{p}$ vectors — one along the [0,1,0] direction and the other along the [1,0,0] direction — should theoretically produce different inclination angles and position angle frequency array, but actually produce a very similar distribution. This shows that although we can recover the θ value relatively accurately, the recovery of the φ value is limited by certain factors. The frequency arrays generated by $\vec{p}$ vectors with different φ values are similar, which means that under the current simulation framework, using the frequency distribution of inclination angle and position angle to restore φ values may not be enough to accurately reflect the real star distribution.

Figure 8



Note. Figure 8 shows a comparison of frequency array distributions generated by mutually perpendicular $\vec{p}$ vectors when the α value is 0.8.

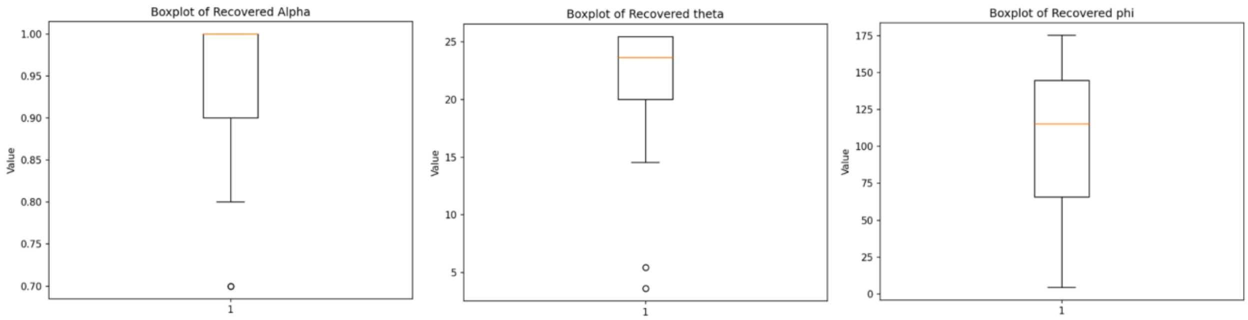
Real data testing

In the actual data testing phase of this project, we used the inclination angle data of 93 Be stars provided by Sigut and Ghafourian as samples (Sigut & Ghafourian, 2023). Our goal is to verify that the simulated database can accurately recover the α values and the θ and φ components of the $\vec{p}$ vector from the real data. The test process includes conducting KS tests on the sample data and the inclination angle frequency data in the database, and recording all results with KS values close to 0, which shows that the sample data is highly similar to the distribution of a certain record in the database.

The results shown in Figure 9 show that the recovered α value is closer to 1, which means that the sample data has a very high weight value. A higher α value implies that there may be a preferred direction in the sample data set that is highly correlated with a specific $\vec{p}$ vector in the simulated database. What is even more noteworthy is that the recovered θ values are mainly concentrated between 15 and 25 degrees. θ values in this range indicate that the direction of the rotation axis tends to be perpendicular to the galactic plane. In astronomy, a θ value of 0 degrees means toward the north galactic pole, while 90 degrees means completely parallel to the galactic plane. Thus, a θ value of 15 to 25 degrees is further evidence that we can extract from real data the orientation of the axis of rotation that tends to be perpendicular to the galactic plane.

However, the recovered φ values show large variability, as shown in the box plot in the figure. This shows that although the recovery of α and θ values is accurate, the recovery accuracy of φ values is still insufficient. This result is consistent with previous simulation tests, suggesting that our method has certain limitations in recovering φ values.

Figure 9



Note. Figure 9: Shows boxplots of α, θ and φ values recovered from 93 actual Be star data provided by Sigut and Ghafourian.

Discussion

Threats to the validity of the results

The validity of research results is affected by a variety of factors, such as assumptions made by simulation methods that may not exactly match real-world conditions. Our model only uses a weight value α and a $\vec{p}$ vector to model the preferred directions that may exist in star clusters. However, the distribution of star rotation axis directions may be more complex than our model. Secondly, the quality and completeness of the data set also have an important impact on the results. Inaccurate or incomplete data can lead to misleading conclusions.

To reduce these threats, we used various statistical methods to test and validate the results, such as KS test and chi-square test to ensure that the findings are based on robust data analysis. By using proven simulation techniques, we increase the confidence in our results. However, residual threats remain, particularly in terms of restoring φ values, which may require further research to address.

Implications of the research results

The findings have potential implications for related research fields in physics and astronomy. For example, by revealing the relationship between the axis of rotation of a Be star and the plane of the Milky Way, we may advance our understanding of the structure and star formation of the Milky Way. These results can improve existing astrophysics models and may also inspire new research directions, such as a deeper exploration of the angular momentum distribution of galaxies.

On the practical side, the results of this study could provide astronomers with new insights when observing and modelling interstellar matter. Understanding the distribution characteristics of stellar rotation axes may be particularly important for developing new observation strategies and data analysis methods.

Limitations of the results

The results of the database tests point to a limitation in our study approach, namely that current simulation methods may not be sufficient to recover all parameters. In particular, when we try to recover specific rotation axis preference directions from the database, we find that θ values can be recovered accurately, while the recovery of φ values is less than satisfactory. This may be due to the failure to fully consider the representativeness of the φ value in the actual data in the construction assumptions of the simulated data set.

In addition, the applicability of the results may be subject to certain conditions. For example, 93 real star data may not be large enough to represent an entire star cluster or galaxy, and the accuracy of the simulation depends on the choice of input parameters, which may not always be available or reliable.

These conditions can be minimized by using larger and broader data sets, or avoided or eliminated by further improving simulation methods. Additionally, the robustness of the results can be enhanced by conducting more real-world validations.

Generalizability of the results

The extent to which our findings apply to other contexts remains to be further verified. While our findings are reasonable in the specific environments considered, whether they can be generalized to different types of stars or galactic structures requires additional research. Further observational studies and more extensive data analysis will help determine the generalizability of these results and potentially extend these concepts to broader astronomical research.

Conclusions

The question addressed in this study is to explore the correlation between the direction of the rotation axis of Be stars and the characteristics of the Milky Way, especially whether they tend to maintain a certain alignment with the galactic plane. The main goal of the research is to use Monte Carlo simulation methods to examine the preference of the Be star rotation axis direction distribution by constructing a simulation data set and using various statistical tests (see Research Objectives).

The simulation data set model shows high accuracy, especially in recovering the θ value of the Be star's rotation axis inclination, but it has limitations in the recovery of the φ value, as shown in Figure 7. These findings are innovative for understanding the distribution of stellar rotation axes within galaxies, and especially reveal that there may be a problem of insufficient sensitivity to certain azimuthal angles under current data analysis methods.

Through Monte Carlo simulation, we can accurately recover the θ component in the direction of the rotation axis of the Be star under high α values (see Methodology). However, there are limitations in the recovery of the φ component (see Actual Data Test), which reveals that under the current data analysis method, there may be a problem of insufficient sensitivity to certain azimuth angles. The novelty of the research lies in the use of modern data science methods to quantitatively analyze long-term questions in astrophysics, and the unique disk of Be stars provides more precise directions.

The simulation database needs to be improved in restoring the complete three-dimensional spatial direction. This points to the need for improvements in simulation methods and statistical testing in future research to increase the comprehensiveness and accuracy of recovering the direction of the star's rotation axis.

Future Work and Lessons Learnt

In future research, we plan to use the position angle data of nearly 500 Be stars provided by Curé for further testing (Curé et al., 2010). With these data, we hope to deepen our understanding of the correlation between Be star rotation axis orientation and galactic features, and to test the applicability and accuracy of our model on a wider data set.

In addition, we plan to expand the observational data set to include more Be star samples and use the latest observational technology to enhance the accuracy and completeness of the data. The improvement of simulation algorithms is also the focus of future work, especially for the accurate recovery of φ values, which may be achieved by introducing more complex physical processes and additional astrophysical variables. The integration of multi-dimensional data will also be an inevitable trend in future research, especially the use of machine learning and deep learning algorithms to process and analyze these data sets.

An important lesson we learned during this research project is that even the most sophisticated simulations can differ from real-world observations due to complex cosmic phenomena. Therefore, rigorous empirical verification of simulation results is indispensable. We also recognize that in astrophysical data analysis, existing statistical methods may not fully capture all relevant physical phenomena, especially when dealing with multidimensional data. These novel lessons are important for advancing scientific research and provide a valuable reference by highlighting aspects that require more careful consideration in future work.

Acknowledgments

I would like to express my sincerest gratitude to my project supervisor, Prof. Aaron Sigut from the Department of Physics and Astronomy. His guidance was instrumental in the planning and structural design of this project. His expertise and insight helped me navigate through the intricate process of this research, and his willingness to answer questions and guide the structural design was invaluable.

Additionally, I am deeply thankful to my course instructor, Prof. Nazim Madhavji from the Department of Computer Science, for providing a comprehensive format guide for the proposal, progress reports, and final report.

Their support and mentorship have been pillars of strength throughout this research journey, and I am profoundly grateful for their contributions.

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